

A Lean 4 formalisation of “ $\zeta(5)$ is irrational” (A. Fauzan, 17 Sep 2026): what it proves and what it assumes

Companion note to `docs/STATUS.md`

24 September 2026

Abstract

The project compiles, and it contains no `sorry`. There is one theorem, `Zeta5.zeta5_irrational : Irrational zeta5`. The paper’s entire route to it is wired up beneath it, with nothing short-circuited: Theorem 2.1, Propositions 2.2, 4.1, 4.3, 5.1, 5.2 and 6.3, Lemma 3.3, and the numbered equations between them. The theorem rests on *two* external axioms, which Mathlib does not have: Hermite’s integral formula for the Hurwitz zeta function, and the prime number theorem in the partial-summation form that the proof consumes. It rests on nothing else.

The paper’s (6.14), the logarithmic-energy bound of §6, was the last `sorry` on 23 September. It is now proved, with its statement unchanged, in the 37 modules of `Zeta5/Sec6/`. Their route differs from the paper’s at several places: a real-line regularisation of the logarithmic kernel instead of circles, a different bound on the smoothing error, and a kernel-checked certified partition of its own in place of Table 2. Every statement that the proof of Theorem 1.1 uses is therefore machine-checked, conditional only on the two axioms.

At three places in §§3–4 the formal proof takes a different route from the paper’s own proof: Lemma 3.2 (the distribution formula, with the far poles), the polynomial part of the proof of Lemma 3.3, and the unimodularity arguments. The paper’s proofs of those three steps are bypassed rather than checked. The assumption list is recomputed from the Lean environment on every build. An adversarial certification on 24 September confirmed it by five independent methods and a full kernel replay. It also showed, by a kernel-level comparison, that no statement or definition of the 23 September state changed.

1 The build

Lean 4.34.0 with Mathlib v4.34.0. There are 69 modules under `Zeta5/` (37 of them in `Zeta5/Sec6/`) plus the root file, 28 918 lines of Lean. With all project build artifacts removed, a rebuild of every `Zeta5` module from source takes about 8 minutes on a 12-core machine (7 min 26 s to 8 min 1 s in four clean builds). It ends with

Build completed successfully (8994 jobs).

with zero errors and *no declaration uses ‘sorry’* warning. The 17 warnings are style lints. Fourteen are unused automatically included section variables, and three are unreferenced hypothesis names. Removing them would change theorem signatures.

2 The one theorem and its assumptions

```
theorem Zeta5.zeta5_irrational : Irrational zeta5
```

with $\text{zeta5} : \mathbb{R} := \sum' v : \mathbb{N}, 1 / ((v : \mathbb{R}) + 1)^5$, i.e. $\zeta(5)$. `Zeta5/Audit.lean` prints, on every build:

```
ASSUMPTION REPORT for Zeta5.zeta5_irrational

(A) standard Lean axioms:
  Classical.choice
  Quot.sound
  propext

(B) explicit external axioms of this project (Zeta5/Axioms.lean):
  Zeta5.Axioms.hermite_pole_integral
  Zeta5.Axioms.pnt_prime_riemann_sum

(C) sorry(s) - unfinished steps of the paper's own argument, 0 in all:
  (none)

(D) any other axiom (must be empty):
  (none)

[self-check: walker agrees with Lean.collectAxioms, 5 axiom(s)]

every sorry of the Zeta5 namespace is used by Zeta5.zeta5_irrational.
```

The last line is printed vacuously: there is no `sorry` at all. `#print axioms Zeta5.zeta5_irrational` gives exactly `[propext, Classical.choice, Quot.sound, Zeta5.Axioms.hermite_pole_integral, Zeta5.Axioms.pnt_prime_riemann_sum]`. Independent scans share no code with `Audit.lean` (`cert/final/`). One covers all 3862 declarations defined in the 70 `Zeta5` modules. Another walks the compiled `.olean` files directly. They find exactly two axioms and no declaration that mentions `sorryAx`.

The ground rules

Anything internal to Fauzan's argument is proved or a `sorry`, never an axiom. Neither axiom mentions any constant or table of the paper, and the only object of the paper that either one mentions is the explicit weight w of (2.10). A finite explicit computation is never an axiom: Appendix B's exact rational integrals and Appendix A's Table-1 evaluations and Table-2 tiling are *proved*. And no statement is changed to make it provable: the statements of the interface theorems and of the 19 leaves of the §6 blueprint were fixed before their proofs were written. A comparison at the level of kernel terms between the state of 23 September (one `sorry`) and the present state finds no change in any statement or definition. Every one of the 3272 declarations of the earlier state is still present, and all 590 new declarations are in `Zeta5/Sec6/` (`docs/CERTIFICATION.md`, §5).

3 (B) The two external axioms

`Zeta5/Axioms.lean` is the only file containing axiom declarations, and it contains two. Each docstring gives the statement, a citation, and why Mathlib lacks it.

`hermite_pole_integral`

$$\int_0^\infty \frac{w(y)}{y^2 + a^2} dy = a^4 \zeta(5, a) - \frac{1}{2a} - \frac{1}{4} \quad (a > 0),$$

with w the weight of (2.10) and $\zeta(5, a) = \sum_{k \geq 0} (k+a)^{-5}$: the display in the middle of p. 5. **Citation:** DLMF 25.11.29, Hermite’s integral formula for the Hurwitz zeta function — the reference the paper itself gives. **Why not Mathlib:** Mathlib has `HurwitzZeta.hurwitzZeta` and its functional equation, but no Hermite/Abel–Plana integral representation and no theory of the Eisenstein-type weight w . It is slightly stronger than DLMF 25.11.29 alone: the paper’s four integrations by parts on p. 5 are absorbed into it. Numerically confirmed from the Lean definition of w at 40 digits.

`pnt_prime_riemann_sum`

For $0 \leq a < b$ and φ bounded on $[a, b]$ and continuous off a finite set,

$$\frac{1}{X} \sum_{\substack{p \text{ prime} \\ aX < p \leq bX}} \varphi(p/X) \log p \longrightarrow \int_a^b \varphi(y) dy \quad (X \rightarrow \infty).$$

Citation: the prime number theorem in the form $\theta(x) \sim x$ (the paper’s [9, §27.12]), carried through Abel summation and the Darboux criterion (Apostol, *Introduction to Analytic Number Theory*, Thms 4.2, 4.4, §4.3). **Why not Mathlib:** as of v4.34.0 Mathlib has Chebyshev-type bounds but no $\theta(x) \sim x$. It is the *partial-summation corollary* of the PNT, the one place where a reader must grant more than a bare citation; its hypotheses carry no information about the paper’s functions, and all of them are discharged inside `PrimeSum.lean`. The certification derives from it in Lean, with no `sorry`, the dyadic form $(\theta(2X) - \theta(X))/X \rightarrow 1$ of the prime number theorem, so it is not vacuous.

Each axiom has exactly one direct user in the dependency cone of 69 102 constants: `Positivity.integral_wt_div_pole` for the first, and `PrimeSum.tendsto_primeSum` for the second. The proof of (6.14) uses Hermite’s axiom through Proposition 2.2’s moment representation, which Andréief’s identity (6.10) needs. `#print axioms Zeta5.RealBound.eq_6_14` gives `[propxt, Classical.choice, Quot.sound, Zeta5.Axioms.hermite_pole_integral]`.

4 (C) Unfinished steps: none. How (6.14) was proved

(6.14), p. 20, states that for $\Delta_K(\zeta(5)) > 0$,

$$\log \Delta_K(\zeta(5)) \leq 2h(h + 6N - K) \log K + (\lambda M_0 - I(\rho))K^2 + 18h \log K + 160h,$$

with $K = 40n$, $N = 3n$, $h = 37n$ and $\lambda = 37/40$. Lean states it with $I(\rho)$ as the closed form (A.2) and $M_0 = -1329/200$. The hypothesis is discharged by `delta_pos` (Proposition 2.2). The statement is the one that was a `sorry` on 23 September. It moved unchanged from `RealBound.lean` to `Sec6/Final.lean`. The route, whose module docstring gives every step:

step	Lean, and how it differs from the paper
(6.10)–(6.13)	Sec6/Gram.lean : Andréief’s identity by a double permutation sum, applied to the moments of Proposition 2.2. $\int_0^\infty (1+y)^5 e^{-y/K} dy \leq 326K^6$ replaces the paper’s 652. The factor $1/h!$ of (6.10) is kept through to (6.14); the paper keeps it in (6.13) and discards it only when taking logarithms. With it, any configuration bound up to $(\lambda M_0 - I(\rho))K^2 + 3h \log K + 131h$ suffices.
(6.6)–(6.9), Lemma 6.2	Sec6/Energy.lean , CND.lean , Frullani.lean , GaussPD.lean , Swap.lean . Deviation : the points stay Dirac masses on the real line, and the kernel is regularised, $k_\varepsilon(x) = \frac{1}{2} \log(x^2 + \varepsilon^2) \geq \log x $ with $\varepsilon = (16K)^{-2}$, instead of the points by circles in $\mathbb{C}$. The zero-mass inequality of Lemma 6.2 is proved for k_ε , from a Frullani representation and positive definiteness of the Gaussian kernel. The result, (6.9’), has $2h \log K + 20h$ where the paper has $(120 + \sqrt{2})h$.
cross term, p. 19	RhoCross.lean , SmoothArc.lean , SmoothErr.lean , SmoothAux.lean , Antideriv.lean . The smoothing error is at most $\pi(\delta + \pi\sqrt{2\delta})$ per arcsine component, so the cross term is at most $88\sqrt{\varepsilon} + 420\varepsilon$. This replaces the paper’s mass bound and $60\sqrt{\varepsilon}$.
(A.1), (A.2), (A.5)	LogCos*.lean , ArcsinePot.lean : (A.1) for every interval, on it from $\int_0^\pi \log \sin = -\pi \log 2$ and off it from Mathlib’s circle averages. RhoEnergy.lean and its leaves: (A.2) is used as $I(\rho) \leq I_k(\rho, \rho)$ for the regularised kernel. Field.lean : the closed form (A.5) of the field V of (6.1).
(6.2), (6.7), (6.8), Lemma 6.1	Sec6/Num/ , Potential.lean . Deviation : the bound $\mathcal{B}(l, r)$ of (A.9) on Table 2’s 684 cells is not used. Instead the closed forms of $2U^\rho - V$ are evaluated with certified rational interval arithmetic on 1049 cells of $[0, 2]$ and 2 cells of $[2, 4]$. Each cell is checked by the kernel (decide +kernel ; there is no native.decide). An analytic bound for $t \geq 4$ replaces (6.8). The partition is generated by numerics/sec6/gen_lean.py .

The 19 leaf statements were fixed in a blueprint commit before any of them was proved, and they did not change afterwards. They were tested numerically first (**numerics/sec6/check_leaves.py**: 133 checks, 7 of them must-fail controls). Sixteen leaves were tested directly, the two branches of (A.1) together, and one bilinear expansion not at all; the kernel checks them all. Downstream, Proposition 6.3 ((6.16) from (6.14) and (6.15)), (7.1), Theorem 2.1 and Theorem 1.1 were already proved.

5 How the four large interface statements were proved

All four are closed in the strict sense: **#print axioms** on each shows no **sorryAx** (verbatim: **'Zeta5.outer_local_analysis'** depends on axioms: **[propxt, Classical.choice, Quot.sound]**, and likewise for the other three). Nothing was moved one level down.

statement	how it was proved
shared lemma HermiteBasis. det_coeffMatrix_ unimodular	<i>A basis of integer polynomials that reduces mod p to a Hermite-interpolation basis $\prod_{c \neq a} (t + u_c)^{m_c} (t + u_a)^i$ with distinct nodes is $\mathbb{Z}_p$-unimodular:</i> linear independence over any field via coprimality of the class factors, a dimension count, and reduction of the determinant mod p . This is p. 10’s “triangular local bases whose diagonal entries are units” and p. 12’s “these form a unimodular basis”, by one argument for both.
outer_local_ analysis, §4.2, (4.10)–(4.12)	OuterBasis.lean (1943 lines), following pp. 11–12 step by step: square classes and the counts $\ell - \delta$ and $\ell - 2$; the rows $P_a q_{a,i}$ of (4.11) and the zero-class rows; $v_p(\mu(t^e)) \geq -1$, with integrality for $e < 2p - 3$; the splitting $G_K = A + p^{-1}L$; $L = UL_0U^\top$ with $L_0[i, j] = 0$ for $i + j < K - 6N + 2p - 3$ in the monomial basis, hence $\text{rank } L \leq r_p$ — p. 12’s reason, applied in the basis where it holds; and the four entry-bound cases behind (4.12).
PrimeSum. eq_5.7_ uniformity, (5.7) and §5.2	Uniformity.lean (1396 lines), with the explicit constant $C = 400M^2$ (the true size is about $4M$). Inner: each class’s contribution is the (5.4) integrand at $z = a/p$, and an elementary Riemann-sum estimate with two breakpoint cells bounds the error; the extras sum exactly; p. 14’s transition remark is the identity $I(T+1) - I(T) = T - x - 2$. $v_p(S_K)$: Legendre with $p^2 > 2K$. Outer: $K T(p/K)$ is an explicit integer and the comparison is linear integer arithmetic in each range.
CrudeBound. crude_entry_ bound, Lemma 3.3 at (3.11)	Lemma33.lean (1155 lines) proves Lemma 3.3 itself in the paper’s form, and the pullback (3.1) for every $\mu_X(B/D_{\text{tail}})$. Pole terms as in the paper; the polynomial part by a different route (§6, item 5).
Section3. entry_bounds_ 4.2.4.3, Prop. 4.1’s basis and entries	InnerTate, InnerGeneral, InnerEntries (2240 lines). B is the Gram matrix in the basis (4.5), $c = (\det U)^{-2}$ a p -adic unit by the shared lemma; the entry bound $\min((4.3), \min_c(4.2))$ comes from a general local bound $v_p^G(\tau_X^{\text{ext}}(W/Q)) \geq \min_a(u_a - t_a) - 4$, proved from Raabe’s multiplication theorem for τ ($m^4\tau(P) = \sum_{a < m} \tau(P(a + mz))$), via Mathlib’s sum_bernoulli) and a per-class estimate using a <i>truncated</i> inverse of the far-pole factor.

Every sub-statement was tested in exact arithmetic before it was proved (scripts in **numerics/**): the (4.11) decomposition at $K = 40, 80, 120$ (0 failures; $\text{rank } L = r_p$ wherever computed; a must-fail control without the $+1/(2j)$ of (2.3) fails 13 times), the inner entry bounds on all 703 entries at $K = 40$ and all 2775 at $K = 80$, Lemma 3.3 on all 703 entries of (3.11) at $K = 40$ for eleven primes, and (5.7) over every prime of the ranges at K up to 520 000.

Consequences. Propositions 4.1, 4.3, 5.1, 5.2 and (3.12) are fully proved (5.2 modulo the PNT axiom). The §4.1 combinatorics (**Section41.lean**, including the repair of the gap in the p. 10 inequality, item 18 below) and the §4.2 local lemmas (**OuterLocal.lean**) are load-bearing: 104/148 and 98/107 of their declarations are in the dependency cone of the main theorem.

6 Faithfulness

Statements. Two kernel-level comparisons show that no statement was changed to make it provable.

- On 23 September, the statements of the four interface theorems proved last, of `eq_6_14` and of `zeta5_irrational` were byte-identical to those recorded in a snapshot taken before the four proofs were written. No statement or definition of that snapshot had changed.
- On 24 September, every declaration of the 23 September state was found to have an unchanged type and value. The only exceptions are 16 compiler-generated auxiliary lemmas, which differ only in hygienic binder names (`docs/CERTIFICATION.md`, §5).

Statements checked against the PDF page images (pp. 5–15, 17–25). `eq_6_14` is (6.14) and `prop_6_3` is (6.16), term for term, with the paper’s constants; Table 1, $I(\rho)$, M_0 and (6.4) were re-derived from the PDF. `Lemma33.lemma_3_3` is Lemma 3.3 as printed on p. 8, for every p , K , d ; its hypothesis “ A integer-valued on $\mathbb{Z}_p$ ” is taken as $v_p(A(z)) \geq 0$ for $z \in \mathbb{Z}$, which is equivalent by density. `pullback` is (3.1) on the whole domain of μ_X . `raabe_tau` is (3.7) for polynomials. The weights of `OuterBasis` are the doubled (4.12) exactly, zero class included. **No weakened statement was found.**

Deviations, all recorded in docstrings:

1. `eq_6_14` is stated with $I(\rho)$ and M_0 as *defined real numbers*: the closed form (A.2) and $-1329/200$, not the logarithmic energy. Its proof connects them to the energy argument through $I(\rho) \leq I_k(\rho, \rho)$ for the regularised kernel and through (6.2)/(6.7) for the closed forms (items 19 and 22). The equality of `Irho` with the singular energy $\iint \log |t - u| d\rho d\rho$ is not stated.
2. `lemma_3_1` proves (3.4) for every *partial* sum $\sum_{j \leq J} p^j U_j$, not for the series in the Tate algebra. It is also *not in the cone*: Proposition 4.1 goes around it (item 7). The same holds for `lemma_4_2` in its abstract-field form; Proposition 4.3 uses the rank form `vanishing_of_rank`.
3. `outer_local_analysis` **states the literal rank form of (4.10)**, $\text{rank } L \leq r_p$. An earlier formulation of this interface statement, as a set of vanishing columns of L in the basis (4.11), was false for the paper’s own L (found in the referee audit of the preprint) and was replaced before the proof was written. The proof shows why: the vanishing is in the monomial basis, exactly as p. 12 argues, and $L = UL_0U^\top$ carries the rank bound over. The statement proved is (4.10) as printed.
4. `AppendixB.eq_5_16_5_17` proves the finite-interval difference form rather than the two improper integrals (5.16) and (5.17) — formally weaker than p. 16, and exactly what is consumed.
5. **Lemma 3.3 is proved by a different route.** The polynomial part does not use the paper’s two-scale estimate $P(\mathbb{Z}_p) \subset p^{-L_0-M_0}\mathbb{Z}_p$, (3.8) or (3.9); it expands A in the binomial basis $\binom{x+K}{k}$ and computes τ of each quotient exactly. The *statement* of Lemma 3.3 is verified; its printed proof is not checked line by line. The $-v_p(24)$ turns out to be unnecessary.
6. `eq_5_7_uniformity` bundles three claims with one constant, proved with $C = 400M^2$.

7. **Lemma 3.2 (3.7) is not formalised as stated.** The Tate-algebra extension, (3.5), the parameter $Y = p^5X + C_p$, and (3.7) for rational functions do not appear. Proposition 4.1 is proved without them: pole terms bounded directly, the polynomial part through Raabe (the polynomial half of (3.7), the paper’s own justification) and a truncated-inverse estimate. Do not say “Lemma 3.2 is formalised”.
8. $b_a < 6\alpha x + 3$ strictly, where the paper prints $\leq$ (towards strength; see item 18).
9. Lemma 4.2 takes $v(\text{pinv}) \geq -1$ rather than literally p^{-1} (more general).
10. (5.21) and (5.16)–(5.18) are stated for $M \in 40\mathbb{Z}_{>0}$, as printed.
11. (5.8)/(5.9) are nested ifs, differing from the paper on a null set.
12. Naming trap: `InnerAlloc.L 0` is not the paper’s $L_0 = 4M + 10$.
13. (6.11) is proved as printed via a sharper intermediate bound.
14. `PrimeSum.RR_bddOn` proves the crude $|R| \leq 200M^2$, not the paper’s sharp bound; nothing needs more.
15. **The p. 12 divided-difference step needs more than the paper says.** $H_{p-a}^{(5)} \equiv H_{a-1}^{(5)} \pmod{p}$ alone leaves a residue $-1/a$ (machine-checked); the proof also uses the $-1/4 + 1/(2j)$ tail of (2.3), and the must-fail control shows it is essential. A point the author may want to revise, not an error in the result.
16. Unimodularity of (4.5) and (4.11) is proved by one mod- p Hermite-basis argument rather than the paper’s two separate justifications.
17. The §4.2 local analysis uses only $p \geq 7$, $K < 3p$, $2K < p^2$, $2N < p$, $5N \leq 2p - 2$ — neither $p \leq K$ nor $K \geq 200M^2$ (towards strength).
18. **A gap on p. 10, repaired.** The paper asserts $b_a \leq 6\alpha x + 3$ without proof, attributing it to (4.4); it is a statement about ℓ_N , equivalent to $p\ell_N(a) \leq 2N + p$. The naive bound $\ell_A(a) \leq 2\lfloor A/p \rfloor + 2$ gives only $b_a \leq 6\alpha x + 6$, which does not make the dimensions of (4.4) provably nonnegative. The inequality is proved in the strict form $p\ell_A(a) < 2A + p$ by the case split on $v_A = A \bmod p$ (`Zeta5.el1_lt_caseSplit`) and independently from the closed form of ℓ_A (`Zeta5.el1_lt`, the one in the dependency cone); Lean checks by `rfl` that the two statements are identical. A point the author may want to revise, not an error in the result.
19. **Kernel regularisation on the real line** instead of circles in $\mathbb{C}$ (§6.1). Lemma 6.2 is proved for k_ε , not for the singular kernel. The result, (6.9’), has $2h \log K + 20h$ in place of $(120 + \sqrt{2})h$.
20. **The smoothing error** is bounded by $\pi(\delta + \pi\sqrt{2\delta})$ per arcsine component, instead of the paper’s mass bound and $60\sqrt{\varepsilon}$. The paper’s constant ≤ 60 is proved but not used.
21. **The Gram step** keeps $1/h!$ through to (6.14) and uses $326K^6$ in place of 652. The paper’s own (6.9) constants are admissible (`paper_6.9_admissible`).
22. **(6.2) is not proved by Appendix A’s route.** (A.9) on Table 2 is replaced by the formalisation’s own certified partition ($1049 + 2$ cells, `decide +kernel`) and an analytic tail for $t \geq 4$. Table 2’s tiling of $[0, 2]$ is proved but unused.
23. **(A.1)** is proved in normalised form and scaled: on $[-1, 1]$ by the product formula for $\cos \varphi - \cos \theta$, and off it by Mathlib’s circle averages.

7 Trusting the audit metaprogram

`Zeta5/Audit.lean` walks the dependency cone and reports the groups above. It avoids two traps that a naive dependency walker falls into: `ConstantInfo.value?` is `none` for theorems in Lean 4.34, and reachability must match `Lean.CollectAxioms.collect`. It checks itself against `Lean.collectAxioms` on every build. All its `#sorry_tree` controls print `depends on NO sorry`, including `eq_6.14`, the negative control of 23 September, and five §6 controls.

An adversarial certification on 24 September (`docs/CERTIFICATION.md`; scripts in `cert/final/`, sharing no code with `Audit.lean`) found that five independent methods agree name for name:

- `#print axioms`;
- a breadth-first walker, which catches planted `sorry`s, including ones hidden in `where` auxiliaries and instance fields;
- a module-keyed scan of all 3862 project declarations;
- `Audit.lean` itself;
- a reader that walks the compiled `.olean` files without Lean’s environment.

The cone has 69 102 constants, with no `sorry` and the two project axioms. A full kernel replay of all 70 modules by `leanchecker` passed (52 minutes). `#print axioms` and `collectAxioms` read axiom lists stored in the `.olean` files, whereas the three walkers traverse the proof terms; all five agree.

Two minor weaknesses of `Audit.lean` were found: its `sorry` list skips internal names, and its orphan line prints even when there is no `sorry`. The independent scans have neither weakness.

8 What to do next

- (1) Reduce `pnt_prime_riemann.sum` to $\theta(x) \sim x$ (Abel summation plus a Darboux sandwich).
- (2) Formalise p. 5’s four integrations by parts, so that `hermite_pole_integral` becomes DLMF 25.11.29 verbatim.
- (3) Points the author may want to revise: the p. 10 inequality (item 18), the p. 12 divided-difference step (item 15), and the phrasing of the rank argument for (4.10) (item 3).

How to report this result. “ $\zeta(5)$ is irrational, machine-checked conditional on two external results entered as axioms — Hermite’s integral formula (in the integrated-by-parts form of p. 5) and the prime number theorem in partial-summation form.” Both axioms are slightly stronger than their bare citations (§3). Nothing internal to Fauzan’s argument is assumed. Never write “modulo standard facts” without naming the two axioms.